7. Which one of the following is probability sampling
   (a) purposive sampling
   (b) judgment sampling
   (c) simple random sampling
   (d) Convenience sampling

8. In simple random sampling from a population of N units, the probability of drawing any unit at the first draw is
   (a) $\frac{n}{N}$ (b) $\frac{1}{N}$ (c) $\frac{N}{n}$ (d) 1

9. In …….. the heterogeneous groups are divided into homogeneous groups.
   (a) Non-probability sample
   (b) a simple random sample
   (c) a stratified random sample
   (d) systematic random sample

10. Errors in sampling are of
    (a) Two types (b) three types
    (c) four types (d) five types

11. The method of obtaining the most likely value of the population parameter using statistic is called ……
    (a) estimation (b) estimator
    (c) biased estimate (d) standard error

12. An estimator is a sample statistic used to estimate a
    (a) population parameter
    (b) biased estimate
    (c) sample size
    (d) census

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13. ……..is a relative property, which states that one estimator is efficient relative to another.
    (a) efficiency (b) sufficiency
    (c) unbiased (d) consistency

14. If probability $P[|\hat{\theta} - \theta| < \varepsilon] \to 1$ as $n \to \infty$, for any positive ε then $\hat{\theta}$ is said to ……..estimator of θ.
    (a) efficient (b) sufficient
    (c) unbiased (d) consistent

15. An estimator is said to be ……. if it contains all the information in the data about the parameter it estimates.
    (a) efficient (b) sufficient
    (c) unbiased (d) consistent

16. An estimate of a population parameter given by two numbers between which the parameter would be expected to lie is called an……..interval estimate of the parameter.
    (a) point estimate
    (b) interval estimation
    (c) standard error
    (d) confidence

17. A _____ is a statement or an assertion about the population parameter.
    (a) hypothesis (b) statistic
    (c) sample (d) census

18. Type I error is
    (a) Accept H_0 when it is true
    (b) Accept H_0 when it is false
    (c) Reject H_0 when it is true
    (d) Reject H_0 when it is false

19. Type II error is
    (a) Accept H_0 when it is wrong
    (b) Accept H_0 when it is true
    (c) Reject H_0 when it is true
    (d) Reject H_0 when it is false

20. The standard error of sample mean is
    (a) $\frac{\sigma}{\sqrt{2n}}$ (b) $\frac{\sigma}{n}$ (c) $\frac{\sigma}{\sqrt{n}}$ (d) $\frac{\sigma^2}{\sqrt{n}}$

Miscellaneous Problems

1. Explain the types of sampling.
2. Write short note on sampling distribution and standard error.
3. Explain the procedures of testing of hypothesis
4. Explain in detail about the test of significance for single mean
5. Determine the standard error of proportion for a random sample of 500 pineapples was

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